THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET B

Chapters 7 & 8: **Proportional Reasoning-1, Fractions in Disguise** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A ratio compares two quantities of the same kind by: 1

(a) addition **(b)** division **(c)** subtraction **(d)** multiplication

2. The ratio 3 : 4 is equivalent to: 1

(a) 9 : 12 **(b)** 7 : 8 **(c)** 4 : 3 **(d)** 6 : 9

3. The ratio of 15 minutes to 2 hours is: 1

(a) 1 : 6 **(b)** 15 : 2 **(c)** 1 : 4 **(d)** 1 : 8

4. If 5 notebooks cost ₹ 120, then 8 notebooks cost: 1

(a) ₹ 160 **(b)** ₹ 240 **(c)** ₹ 192 **(d)** ₹ 200

5. If x and y are inversely proportional and x doubles, y: 1

(a) becomes zero **(b)** becomes half **(c)** stays same **(d)** doubles

6. The marked price is ₹ 2400 and the discount is 25%. The selling price is: 1

(a) ₹ 600 **(b)** ₹ 3000 **(c)** ₹ 2375 **(d)** ₹ 1800

7. The simple interest on ₹ 6000 at 12% per annum for 2 year(s) is: 1

(a) ₹ 2880 **(b)** ₹ 7440 **(c)** ₹ 1440 **(d)** ₹ 720

8. 0.45 as a percentage is: 1

(a) 4.5% **(b)** 45% **(c)** 450% **(d)** 0.45%

9. If the price of an item is increased by 10% and then decreased by 10%, the final price is: 1

(a) the same **(b)** less than the original

(c) more than the original **(d)** double

10. If 20% of a number is 45, the number is: 1

(a) 180 **(b)** 270 **(c)** 90 **(d)** 225

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State two ways to check that two ratios are equal. 2

12. Write the ratio of 2 hours to 40 minutes. 2

13. What is compound interest? 2

14. A town's population is 40000 and it grows 5% in a year. Find the new population. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. The ratio of boys to girls in a class is 7 : 5. If total is 36, find the number of each. 3

16. At what total amount will ₹ 9000 grow in 5 years at 4% simple interest? 3

17. A rope 36 m long is cut into three pieces in the ratio 2 : 3 : 4. Find the pieces. 3

OR

The speeds of two trains are in the ratio 4 : 5. If the second travels 300 km in a given time, how far does the first go? 3

18. At what total amount will ₹ 3500 grow in 2 years at 12% simple interest? 3

OR

At what total amount will ₹ 2400 grow in 4 years at 5% simple interest? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Divide 156 into parts proportional to $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$. 5
(ii) Which part is greatest?

OR

(i) Express ratio 0.75 : 1.5 and 2/3 : 5/6 in simplest form. 5
(ii) Which is greater?

20. (i) Find the principal that amounts to ₹ 5832 in 2 years at 8% CI. 5
(ii) State what is compounding.

OR

(i) In a class, 60% of 40 students are girls. Of these, 25% have bicycles. How many girls have bicycles? 5
(ii) What percent of the class are they?

END OF QUESTION PAPER — Total: 40 Marks

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